Jules Fowler

NSF Graduate Research & UC President's Dissertation Year Fellow

they · them · theirs

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Summary

Astronomer : final year PhD student working in high contrast imaging and extreme adaptive optics.

Software Engineer : `Python` and `Git` expert, Linux enthusiast.

Instructor : Software Carpentry Instructor and Python training developer.

Activist : Queer organizer and UAW-4811 union steward.

Research Experience

Lab for Adaptive Optics (LAO)
Santa Cruz, CA

University of California, Santa Cruz — *Sept. 2020 - present*

- **Comparing predictive wavefront control methods** in simulation and on the SEAL (Santa Cruz Extreme AO Laboratory) testbed.
- Characterizing the performance of **visible light adaptive optics** at Keck Observatory with the ORKID instrument.
- Searching for the **polarized signatures of substellar companions** with the Subaru telescope.
- Mentoring an undergraduate student on phase diverse focal-plane wavefront sensing.
- Advisor : Rebecca Jensen-Clem

Russell B. Makidon Optics Lab
Baltimore, MD

Space Telescope Science Institute (STScI) — *Oct. 2018 - Sept. 2020*

- **Principal Investigator** of the Generalized Lab Architecture for Restructured Optical Experiment (**GLARE**) – an STScI Director's Discretionary Research project to build **autonomous experiment software and hardware controllers** for optics experiments.
- Wrote **fully-automated target acquisition** and `Python` **control software for nPoint tip/tilt controller** as part of an optics testbed for HWO-like coronagraphy experiments.
- Advisors : Rémi Soummer & Marshall Perrin

Exoplanet Characterization Toolkit
Baltimore, MD

STScI — *May 2017 - Jan. 2020*

- **Developed first on-the-fly observation planning tool for exoplanet transits with JWST** (the James Webb Space Telescope).
- Built an `HDF5` database and web interface for grid of forward model atmospheric transmission spectra.
- Advisors : Kevin Stevenson & Nikole Lewis

Space Telescope Advanced Research Group for the Atmospheres of Transiting Exoplanets
Baltimore, MD

STScI — *Oct. 2017 - Oct. 2018*

- Wrote open source **pipeline to reduce WFC3/IR spatial scan slitless spectroscopy data**, fit transit parameters with Markov-Chain Monte Carlo (MCMC), calculate Bayesian statistics, and **produce transmission spectra**.
- Advisors : Hannah Wakeford & Natasha Batalha

Wide Field Camera Three Instrument Team
Baltimore, MD

STScI — *Aug. 2016 - Jan. 2020*

- **Lead of the WFC3 Quicklook project**, developing automated monitors and calibration software on a large-scale web and database application.
- **Principal Investigator of eleven HST calibration proposals**.
- Created user guides and tools for WFC3/IR slitless spectroscopy data reduction and anomaly tracking.
- Advisors : Elena Sabbi & Sylvia Baggett

Tufts University
Somerville, MA

Summer Scholar & Senior Thesis — *May 2015 - July 2016*

- **Compared energy contribution of low energy radio jets to gas outflows in AGN**.
- Used `slurm` parallel cluster computing and `CASA` to **reduce a terabyte of multi-band Very Large Array (VLA) data**.
- Received High Honors for senior thesis detailing analysis and small numbers statistics.
- Advisor : Anna Sajina

University of Texas, Austin
Austin, TX

Research Intern — *May 2012 - Aug. 2013*

- Performed **stellar evolution and structure research on Betelgeuese** and Wolf-Rayet stars.
- Created computational models using `MESA`.
- Advisor : Craig Wheeler

Education

University of California, Santa Cruz — *Santa Cruz, CA*

FINAL YEAR PHD CANDIDATE IN ASTRONOMY & ASTROPHYSICS — *Sept. 2020-present*

- **Graduate coursework:** Astrophysics I & II, Stars and Planets I & II, Galaxies and Cosmology I, High Energy Astrophysics, Observational Astronomy, Adaptive Optics, Digital Control Theory, Numerical Linear Algebra.
- **Advanced to Candidacy:** August 2023

Tufts University — *Somerville, MD*

B.S. ASTROPHYSICS & PHILOSOPHY — *Sept. 2012 - May 2016*

- Graduated with High Thesis Honors.
- Dean's List 2013, 2015

Work Experience

Space Telescope Science Institute — *Baltimore, MD*

SCIENCE SOFTWARE ENGINEER — *Aug. 2016 - Sept. 2020*

- Member of the Wide Field Camera Three Instrument Team and involved with multiple science projects across the institute.
- **Started SpaceGAYs, the first LGBT+ affinity group at STScI**, and member of the Invision Diversity Working Group.
- **Designed Advanced `Python` training** with outside vendor, and gave Software Carpentry inspired introductory `Python` and `Git` training.

Tufts University, Department of Physics and Astronomy — *Somerville, MA*

TEACHING ASSISTANT & TUTOR — *Sept. 2014- July 2016*

- Lab TA for introductory mechanics and class TA for general audience astronomy seminar.
- Tutored individual students and led study groups for introductory physics, calculus, and linear algebra.
- Assisted students with accessible exam accommodations.

Grants & Awards

GRANTS

2026	**UC Santa Cruz President's Dissertation Year Fellowship**, 1 quarter fellowship to support thesis work	*Santa Cruz, CA*
2025	**National Science Foundation INTERN Grant**, supplemental funding for a 2 month internship at the Keck Observatory	*Waimea, HI*
2024	**Subaru Telescope Proposal: Three Little PIGSS**, awarded 3 half nights of 8-meter telescope time in the 24B semester	*Hilo, HI*
2021	**Exoplanet Exploration Program "ExoExplorers" Inaugural Cohort**, seminar series and scholarship to develop early career exoplanet scientists	*Pasadena, CA*
2021	**University of California Osterbrock Mini-Grant**, small project grant to build a coronagraph in a box teaching demo	*Santa Cruz, CA*
2020	**National Science Foundation Graduate Research Fellowship**, 3 year NSF fellowship for exemplary graduate students in STEM	
2020	**University of California Regents Fellowship**, 1 year University of California wide fellowship for graduate students	*Santa Cruz, CA*
2019	**STScI Data Science Innovation Initiative Grant**, Director's Discretionary Research Fund call for projects with a focus on data science and interesting computational problems	*Baltimore, MD*
2015	**Tufts Summer Scholar Grant**, call for summer long independent research projects	*Somerville, MA*
2015	**Tufts Center for STEM Diversity Grant**, grant to create the first collegiate queer science speaker series	*Somerville, MA*

AWARDS

2019	**STScI Achievement Award**, team award for teaching `GitHub` and `Python` to colleagues at STScI	*Baltimore, MA*
2019	**STScI Achievement Award**, team award for contributing to the Exoplanet Characterization Tool Kit	*Baltimore, MA*
2016	**Tufts University Pride on the Hill**, for contributions to LGBT+ climate	*Somerville, MA*
2015	**Tufts Multicultural Service Award**, for contributions to diversity efforts	*Somerville, MA*

Certifications & Skills

Certified SCRUM Master	SCRUM Alliance
Certified Software Carpentry Instructor	Software Carpentry
Laser Safety Certification	Laser Institute of America
`Python, Git, conda, bash, Linux, Mac OS, Windows, LaTeX`	expert
`Mathematica, Markdown, HTML`	advanced
`C, MATLAB, Julia, IDL, IRAF/PyRAF, JacaScript, CSS`	beginner

Service and Outreach

2025	**Big Island Astronomy LGBT+ Pau Hana**, organizer	*Hilo, HI*
2025	**Akamai Mentor Workshop**, workshop participant	*Hilo, HI*
2025	**UCSC Astronomy & Astrophysics Graduate Admissions**, graduate student representative	*Santa Cruz, CA*
2024	**Exocast-73b: Using Adaptive Optics to Find Exoplanets with Jules Fowler**, popular science podcast	
2024	**UCSC Astronomy & Astrophysics Faculty hiring committee**, graduate student representative	*Santa Cruz, CA*
2023	**UCSC Astronomy & Astrophysics Graduate Students**, head grad and faculty liaison	*Santa Cruz, CA*
2022-2023	**UCSC Graduate Mentoring**, mentor to junior astronomy graduate student	*Santa Cruz, CA*
2021-2025	**United Auto Workers 4811 Union Steward**, representative for astronomy graduate students and postdocs	*Santa Cruz, CA*
2021	**NASA Review Panel**, executive secretary	
2019-2021	**AAS Committee for Sexual and Gender Minorities in Astronomy (SGMA)**, member	
2019	**Inclusive Astronomy II**, member of the science organizing committee	*Baltimore, MD*
2019	**STScI/AURA Exoplanet Astronomer hiring committee**, member	*Baltimore, MD*
2019	**STScI Information Technology Head hiring committee**, member	*Baltimore, MD*
2017-2019	**STScI Invision Diversity Working Group**, member	
2015-2016	**out in STEM (oSTEM) @ Tufts**, president and founder	*Somerville, MA*
2015	**Tufts Society of Physics Students** , president	*Somerville, MA*
2014-2016	**Tufts Queer Students Association**, treasurer	*Somerville, MA*

Teaching

UCSC Real Data Games & Astronomy — *Santa Cruz, CA*
GUEST LECTURER — *Aug. 2025*
- Gave a one hour guest lecture on physically accurate simulations in science fiction for undergraduate computational media course.

Center for Adaptive Optics Summer School — *Santa Cruz, CA*
LAB TA — *Aug. 2023 & 2024*
- Assisted with optical simulation lab for week-long summer school in adaptive optics.

UCSC Lamat Coding Bootcamp — *Santa Cruz, CA*
LEAD INSTRUCTOR — *Winter 2023*
- Developed material for and taught `Python` and `bash` workshop for incoming Lamat interns.

UCSC Computational Physics — *Santa Cruz, CA*
CLASS TA — *Winter 2022*
- Led recitation sections and assisted during lecture for upper-division for-majors undergraduate computational physics course.

Software Carpentry at the AAS — *remote*
LEAD INSTRUCTOR — *Jan. 2021*
- Astronomy focused `Python` and `bash` at the 237TH AAS meeting.

Everything You Needed to Know about JWST Exoplanet Transit Data : How to Plan, Reduce, and Fit Your Data with the Exoplanet Characterization Toolkit Workshop — *Honolulu, HI*
INSTRUCTOR — *January, 2020*
- Submitted proposal for and led an ExoCTK workshop at the 235th AAS meeting.

We Brought a Coronagraph to a Bar! — *Baltimore, MD*
PRESENTER — *June 2019*
- Astronomy on Tap talk with interactive Lyot coronagraph demo.

Software Carpentry at the New York Academy of Sciences — *New York, NY*
INSTRUCTOR — *Jan. 2019*
- Taught general science focused `Python` and `Git` at NYAS.

Software Carpentry at the AAS — *Seattle, WA*
INSTRUCTOR — *Jan. 2019*
- Astronomy focused `Python` and `bash` at the 233RD AAS meeting.

Software Carpentry Inspired Workshops — *Baltimore, MD*
INSTRUCTOR — *Sept. 2018 - June 2020*
- Taught hundreds of students in series of STScI `Python`, `Git`, and `bash` workshops

Tufts University Concepts of the Cosmos *Somerville, MA*
Class TA *Spring 2016*
- Class TA for the introductory/general education astronomy seminar.

Tufts University Physics 1 & Physics 11 *Somerville, MA*
Lab TA *Fall 2015*
- Lab TA for the introductory mechanics courses with and without calculus.

Presentations

Invited

A Keck-cophony of speckles: supporting the next generation of Keck adaptive optics with predictive wavefront control and visible light AO *remote*
Astralis Instrumentation Consortium Seminar *Nov. 2024*
University of Sydney

Machine Learning Approaches to Predictive Wavefront Control *Leiden, NL*
Optimal Exoplanet Imaging Workshop *Feb. 2023*
Leiden University Lorentz Center

Don't Heckle My Speckle: a coronagraph design study for the SEAL testbed *remote*
ExoExplorers Seminar Series *May 2021*
Jet Propulsion Labs Exoplanet Exploration Program

Selected Conference Talks

Characterizing the performance of Keck-II AO in visible light with the ORKID instrument *Lake Arrowhead, CA*
Fall Retreat *Nov. 2024*
Center for Adaptive Optics

Why Predictive Wavefront Control can Help you Better Characterize Exoplanets *Santa Cruz, CA*
Other Worlds Laboratory Summer Workshop *July 2024*
University of California, Santa Cruz

Tempestas Ex Machina: Machine learning approaches to wavefront control *San Diego, CA*
Optics and Photonics: Techniques and Instrumentation for Detection of Exoplanets XI *Aug. 2023*
Society of Photo-Optical Instrumentation Engineers

Standardizing Exoplanet Analysis with the Exoplanet Characterization Tool Kit (ExoCTK) *Denver, CO*
Winter Meeting #232 *June 2018*
American Astronomical Society

Seminars

Ground Control to Major Time-Lag:Using predictive wavefront control to tackle bandwidth errors and improve high contrast imaging *Hilo, HI*
Tech Talk *May 2025*
Institute for Astronomy, Hilo

Big Plans for Bigger Planets: Supporting the next generation of Keck adaptive optics with predictive wavefront control and visible light AO *Santa Cruz, CA*
Planetary Lunch Seminar *Feb. 2025*
University of California, Santa Cruz

I Predict You Will Love This Talk: Why predictive wavefront control can help you dig darker holes and better characterize exoplanets *Pasadena, CA*
Exoplanet Technology Lab Group Meeting *Nov. 2024*
California Institute of Technology

Keck-cophony of projects: predictive control and visible light AO with ORKID *remote*
Adaptive Optics Department Meeting *Aug. 2024*
Keck Observatory

Comparing Data and Model-Driven Predictive Control *Santa Barbara, CA*
Exoplanet Group Meeting *Nov. 2022*
University of California, Santa Barbara

Battle of the Predictive Wavefront Controls: Comparing Data and Model-Driven Predictive Control for High Contrast Imaging *Santa Cruz, CA*

Friday Lunchtime Astrophysics Seminar *May 2022*

University of California, Santa Cruz

Data and model-driven predictive wavefront control for High Contrast Imaging *Nice, FR*

Physical Methods for Observation seminar *Dec. 2021*

Observatoire de la Côte d'Azur

Publications

Journal Publications

ORCAS Keck Instrument Demonstrator

Peretz, Eliad; Wizinowich, Peter; Millar-Blanchaer, Maxwell; et al. (inc **Fowler, J.**)

Journal of Astronomical Telescopes, Instruments, and Systems, Volume 11, id. 014004 (2025)

Chasing rainbows and ocean glints: Inner working angle constraints for the Habitable Worlds Observatory

Vaughan, Sophia R.; Gebhard, Timothy D.; Bott, Kimberly; et al. (inc **Fowler, J.**)

Monthly Notices of the Royal Astronomical Society, Volume 524, Issue 4, pp.5477-5485 (2023)

Implementation of a dark zone maintenance algorithm for speckle drift correction in a high contrast space coronagraph

Redmond, Susan F.; Pogorelyuk, Leonid; Pueyo, Laurent; Por, Emiel; Noss, James; Will, Scott D.; Laginja, Iva ; Brooks, Keira; Maclay, Matthew; **Fowler, J.**; Jeremy Kasdin, N.; Perrin, Marshall D.; Soummer, Rémi

Journal of Astronomical Telescopes, Instruments, and Systems, Volume 8, id. 035001 (2022)

Low-order wavefront control using a Zernike sensor through Lyot coronagraphs for exoplanet imaging. Blind stabilization of an image dark hole

Pourcelot, R.; N'Diaye, M.; Por, E. H.; Laginja, I.; et al. (inc. **Fowler, J.**)

Astronomy & Astrophysics, Volume 663, id.A49, 15 pp. (2022)

A comprehensive analysis of WASP-17b's transmission spectrum from space-based observations

Alderson, L.; Wakeford, H. R.; MacDonald, R. J.; Lewis, N. K.; May, E. M.; Grant, D. ; Sing, D. K.; Stevenson, K. B.; **Fowler, J.**; Goyal, J.; Batalha, N. E.; Kataria, T.

Monthly Notices of the Royal Astronomical Society, Volume 512, Issue 3, pp.4185-4209 (2022)

Predictive wavefront control on Keck II adaptive optics bench: on-sky coronagraphic results

van Kooten, Maaike A. M.; Jensen-Clem, Rebecca; Cetre, Sylvain; Ragland, Sam; Bond, Charlotte Z.; **Fowler, J.**; Wizinowich, Peter

Journal of Astronomical Telescopes, Instruments, and Systems, Volume 8, id. 029006 (2022)

Disentangling the Planet from the Star in Late-Type M Dwarfs: A Case Study of TRAPPIST-1g

Wakeford, H.; Lewis, N.; **Fowler, J.**; Bruno, G.; Wilson, T.; Moran, S. E.; Valenti, J.; Batalha, N.; Filippazzo, J.; Bourrier, V.; Hörst, S.; Lederer, S. M.; de Wit, J.

The Astronomical Journal, Volume 157, Issue 1, article id. 11, 14 pp. (2019)

The Betelgeuse Project: constraints from rotation

Wheeler, J.; Nance, S.; Diaz, M.; Smith, S.; Hickey, J.; Zhou, L.; Koutoulaki, M.; Sullivan, J.; **Fowler, J.**

Monthly Notices of the Royal Astronomical Society, Volume 465, Issue 3, p.2654-2661, (2017)

First Author Conference Proceedings

The future looks dark: improving high contrast imaging with hyper-parameter optimization for data-driven predictive wavefront control

Fowler, J.; Jensen-Clem, Rebecca ; van Kooten, Maaike A. M. ; Chambouleyron, Vincent ; Cetre, Sylvain

Proceedings of the SPIE, Volume 13097, id. 130977N 10 pp. (2024)

Closed-Loop Until Further Notice: Comparing Predictive Control Methods in Closed-Loop

Fowler, J.; van Kooten, M. A. M. ; Jensen-Clem, R.

Adaptive Optics for Extremely Large Telescopes 7th Edition 9 pp. (2023)

Tempestas ex machina: A review of machine learning methods for wavefront control
Fowler, J. & Landman, Rico
Proceedings of the SPIE, Volume 12680, id. 126800E 15 pp. (2023)

Visible extreme adaptive optics on extremely large telescopes: Towards detecting oxygen in Proxima Centauri b and analogs
Fowler, J.; Haffert, Sebastiaan Y. ; van Kooten, Maaike A. M. ; et al.
Proceedings of the SPIE, Volume 12680, id. 126801U 11 pp. (2023)

Battle of the predictive wavefront controls: comparing data and model-driven predictive control for high contrast imaging
Fowler, J. ;Van Kooten, M. A. M. ; Jensen-Clem, R.
Proceedings of the SPIE, Volume 12185, id. 1218582 15 pp. (2022)

A design study for adaptive primary mirrors in 1-2 meter class telescopes
Fowler, J. ; Bowens-Rubin, Rachel ; Hinz, Philip M.
Proceedings of the SPIE, Volume 11820, id. 118200Y 13 pp. (2021)

Technical Reports and Other Publications

In Violation of the Prime Directive: Simulating detriments to Delta-Quadrant civilizations from the starship Voyager's impact on planetary rings
Fowler, J.; Murray-Clay, Ruth
April Fools arXiv (2024)

Wavefront Sensing and Control with the Many Headed Hydra Modal Basis
Fowler, J. ; Deno Stelter, R.
April Fools arXiv (2022)

Analyzing Eight Years of Transiting Exoplanet Observations Using WFC3's Spatial Scan Monitor
Stevenson, K.; **Fowler, J.**
Instrument Science Report WFC3 2019-12, 16 pages

WFC3/UVIS Gain Stability Results for Cycles 24 and 25
Fowler, J.
Instrument Science Report WFC3 2018-17, 24 pages

The Cosmic Ray That Wouldn't Quit: Correcting Cosmic Rays in Overscan Pixels
Fowler, J.
Instrument Science Report WFC3 2017-12, 7 pages

Monitoring the WFC3/UVIS Relative Gain with Internal Flatfields
Fowler, J.; Baggett, S.
Instrument Science Report WFC3 2017-08, 17 pages

Analysis of Dragon's Breath and Scattered Light Detector Anomalies on WFC3/UVIS
Fowler, J.; Markwardt, L.; Bourque, M.; Anderson, J.
Instrument Science Report WFC3 2017-02 (v.1), 14 pages